

Collaborative Subgraph Learning based Spectrum Sensing under Partial Observations

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Abstract

Data-driven spectrum sensing is a key technology for addressing complex challenges in Cognitive Radio Networks (CRNs). Traditional methods are typically designed for simple single-band scenarios and perform poorly in practical wideband applications. In real-world systems, a single Secondary User (SU) is often restricted by energy, time, and hardware capabilities during real-time sensing. Consequently, only local and fragmented frequency information can be obtained. This partial sensing leads to a severe lack of training data. Additionally, the lack of historical records for emerging frequency bands, combined with data incompleteness due to resource constraints, creates training bottlenecks for data-driven models and limits the reliability of sensing. To address these challenges, this paper proposes a novel framework based on Collaborative Subgraph Learning and Hyperbolic Graph Neural Networks (GNNs). This approach enables Secondary Users to perform collaborative sensing through distributed subgraph learning. By utilizing GNNs to extract features and model multi-band correlations, a new distributed GNNs architecture is designed to efficiently detect wideband spectrum occupancy, even with partial observations. Within this framework, all frequency bands in the wideband spectrum pool are treated as a unified graph, while the bands observed by each SU form a subgraph. Subsequently, the complete spectrum graph is constructed through the joint training and aggregation of these subgraphs. By integrating hyperbolic geometry into GNNs, this method better captures the hierarchical structure of spectrum patterns, providing a more accurate and efficient sensing model. Experimental results demonstrate that, compared to the second-best HCNNs model, the proposed framework improves sensing accuracy by 3.8% on average across various test environments, while reducing key resource consumption by 18.4% on average.

CCS Concepts

• **Computing methodologies** → **Feature selection.**

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Keywords

Spectrum Sensing, Cognitive Radio Networks, Collaborative Subgraph Learning, Graph Neural Networks, Spectrum Occupancy Detection

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1 Introduction

Intelligent radio frequency spectrum sensing is a necessary measure to address the scarcity of spectrum resources in the field of wireless communication, and cognitive radio (CR) is widely recognized as one of the key technologies for achieving intelligent spectrum management [22, 23]. Spectrum sensing aims to accurately and efficiently detect spectrum occupancy and improve the utilization of spectrum resources, making it a core component of CR technology.

In signal model matching, traditional spectrum sensing methods based on signal processing techniques, such as energy detection [15], matched filtering [37], and cyclostationary detectors [16], can achieve good performance under the fundamental assumption of fully known signal and channel models. In practical applications, model mismatch caused by factors such as noise and channel uncertainties leads to a significant degradation in the performance of traditional methods [21]. To address this issue, data-driven spectrum sensing methods based on deep learning have been developed in recent years for both single-band [4, 28, 30] and multi-band sensing [8, 29, 33]. These methods have achieved strong sensing performance by leveraging the powerful representational capacity of deep neural networks to implicitly capture underlying structures and patterns from training data.

Although these pioneering methods eliminate the reliance on predefined prior signal knowledge, enhancing model adaptability to complex environments, they still operate under the fundamental assumption of homogeneous settings and idealized observation conditions. But in realistic CR practice, this assumption is often violated due to the limitation that a single secondary user (SU) can only observe a locally portion of the wideband spectrum [26]. One reason is that the limited sensing capabilities and hardware constraints of individual SUs. These bottlenecks restrict each SU

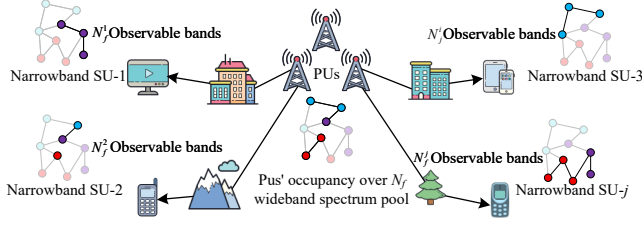


Figure 1: Cooperative sensing for wideband spectrum with partial observations

to only observing limited narrowband part of wideband spectrum pool, as illustrated in Figure 1. As a result, localized narrowband data fails to comprehensively characterize the dynamic occupancy patterns of the entire wideband spectrum [26]. The another reason is that the heterogeneous deployment environments of distributed SUs [12], such as geographical disparities and channel fading variations, exacerbate spatiotemporal inconsistencies in observational data, rendering standalone learning models trained on single-SU data inadequate for generalizing to global spectrum states. To address partial observation limitations, existing works explore collaborative sensing mechanisms to integrate multi-SU data. For instance, centralized methods [11] aggregate raw observation from all SUs at a central server for joint decision-making but suffer from exponentially growing communication overhead and privacy leakage risks. While federated learning frameworks [35] mitigate raw data exchange by sharing model parameters, their homogeneous model assumptions require all SUs to adopt identical neural network architectures and learning tasks, which conflicts with the data heterogeneity caused by partial observations. Therefore, it is imperative to explore collaborative strategy under partial observations for practical CR networks.

To leverage the narrowband sensing data from distributed SUs to train wideband spectrum sensing models and effectively capture the multi-band correlations in wideband spectrum data to improve model performance, this paper proposes a spectrum sensing framework based on heterogeneous graph neural networks. The main contributions of this paper are as follows:

- This paper proposes a new spectrum sensing framework that adopts a hypergraph neural network to model the multi-band correlation and hierarchical structure in spectrum data. By mapping spectrum data to a hyperbolic geometric space, this framework can more effectively capture the hierarchical characteristics of spectrum occupancy patterns. Compared with traditional Euclidean methods, it provides higher accuracy and robustness in environments with partial observations and low signal-to-noise ratio (SNR).
- To address the problem of partial observations and enable collaborative learning among multiple SUs, this paper proposes a collaborative subgraph learning method. Each SU represents its local band data as a narrowband spectrum subgraph, and then constructs a complete wideband spectrum graph through joint training and gradient sharing. This method avoids direct sharing of raw data and leverages the local information of each SU for global model learning, thereby

significantly improving the performance of wideband spectrum occupancy detection.

- To tackle the challenge of insufficient wideband spectrum data, this paper proposes a data generation method. By simulating different modulation schemes and channel allocation strategies, a wideband spectrum dataset that conforms to partial observation conditions is constructed. This method can generate more training samples, alleviate the problem of data scarcity, and enhance the generalization ability of the model in low-data environments.

2 Related Works

Traditional signal processing techniques in spectrum sensing primarily rely on statistical analysis and spectrum feature extraction. However, these methods have limited adaptability to environmental changes and perform poorly under low SNR or strong interference conditions [2, 10]. Furthermore, traditional signal processing methods require explicit prior knowledge and fine-tuned parameter adjustments, making them difficult to handle in complex, dynamic spectrum environments [7, 18, 36]. With the rapid development of deep learning, spectrum sensing has gradually shifted towards data-driven approaches. Deep learning technologies leverage the powerful feature extraction and nonlinear modeling capabilities of neural networks, enabling direct learning of spectrum occupancy patterns from large amounts of spectrum data [9, 19].

Most existing deep learning-based spectrum sensing methods are based on Convolutional Neural Networks (CNNs), which are widely applied in computer vision and natural language processing fields [1]. CNNs learn the geometric correlations of adjacent signal components in the frequency or time domain within the receptive field through convolutional kernels, enabling them to capture local spectral features and the temporal patterns of signals. These convolutional kernels exhibit good adaptability to the translational invariance of spectrum data, allowing them to effectively capture features within a frequency band for modulation recognition [39] and occupancy detection [24] in narrowband sensing scenarios. J. Xie et al. proposed a spectrum sensing algorithm based on activity pattern modeling, APASS, which improves detection accuracy by modeling primary user (PU) activity patterns [20]. However, in real-world scenarios, CR devices are typically mobile or embedded in environments, limiting their performance in high-computation tasks. The devices' computational capabilities may be insufficient to support complex deep learning models, potentially affecting the effectiveness of spectrum sensing. To address these practical issues, researchers must urgently develop more efficient algorithms and models that reduce resource dependence while ensuring the accuracy and reliability of spectrum sensing.

Data parallelism is an important strategy in distributed learning, aimed at leveraging the parallel computing capabilities of distributed nodes in a multi-agent system to efficiently process large-scale datasets [17, 25]. Federated learning and parameter servers are typical distributed training methods that can construct a global task model from distributed data [3]. Although these training methods perform excellently in terms of communication efficiency and privacy protection, they are typically based on the assumption that

all distributed nodes have homogeneous deep neural network structures and that the data is independent and identically distributed (IID), allowing them to collaboratively complete a learning task. This assumption requires an idealized CR scenario, where all SUs must have full observations of the entire wideband spectrum pool. However, in practical CR sensing systems, this condition is often not met. In cases of local observation, local spectrum data and corresponding model structures exhibit heterogeneity. GNNs provide an effective framework for addressing this collaboration issue by representing spectrum data as graph structures, where the local spectrum data of each secondary user can be viewed as a subgraph, and the entire spectrum pool as the full graph. GNNs can flexibly handle these heterogeneous data, facilitating collaborative learning for spectrum sensing tasks, improving the accuracy of spectrum sensing, and eliminating the need for global observations.

In recent years, GNNs have demonstrated significant advantages in handling non-Euclidean data [5, 38]. By leveraging the graph structure to model complex relationships in the data, GNNs can effectively propagate information and aggregate features through edges between nodes [13, 14], thereby capturing correlations between node features, edge information, and the topological structure. In cognitive radio research, graph structures have been used to model multi-band correlations. Some studies abstract frequency bands as nodes and compute specific correlation coefficients between bands to construct edges between nodes. For example, Yang et al. proposed a spectrum prediction method combining graph convolutional networks and recurrent neural networks [29]. Han Zhang et al. proposed the MFFGCN for radio map estimation, which mitigates sparse/uneven data issues and achieves state-of-the-art performance on open datasets [32].

However, existing methods typically construct graphs by calculating multi-band correlations and building adjacency matrices, overlooking the heterogeneity of multi-band correlations caused by different factors [6, 27]. Furthermore, current research mainly focuses on directly constructing the full spectrum pool graph for cognitive radio tasks, which faces feasibility challenges in practical spectrum sensing environments [31, 34]. To address these shortcomings, this study proposes a distributed subgraph collaborative training method to construct a global graph model, thereby more effectively capturing the complex multi-band correlations while enhancing the model's applicability in practical spectrum sensing. To the best of current knowledge, there has been no research on wideband spectrum sensing based on collaborative subgraph learning.

3 Problem Formulation

3.1 Signal Model

First, this paper formalizes the power spectral density (PSD) data collected by distributed SUs in order to construct a signal model under partial observation conditions. As shown in Figure 1, this study divides the wideband spectrum pool evenly into N_f frequency bands, using a Boolean function $y = [y_1, y_2, \dots, y_{N_f}] \in \{0, 1\}^{N_f}$ to represent the occupancy of these frequency bands by the primary user (PU). The channel gain is calculated through path loss and

shadow fading, as shown in the following formula:

$$h_{j,n} = \beta \left(\frac{d_0}{d_{j,n}} \right)^\alpha 10^{\frac{X_\sigma}{10}}, \quad (1)$$

where β is a constant related to antenna characteristics and average attenuation, α is the path loss exponent, $d_{j,n}$ is the distance between SU- j and the PU on the n -th frequency band, d_0 is the reference distance, and X_σ is a normally distributed random variable with a mean of zero and a standard deviation of σ , used to model the shadowing effect that reflects the channel shadow fading between the PU and SU- j on the n -th frequency band.

The value of y_n is 1 if the n -th frequency band is occupied by the PU, and 0 if it is unoccupied. Assume that the cognitive radio system has J secondary users, where SU- j ($\forall j \in [1, J]$) receives the radio signal's power spectral density observations, which, after appropriate sampling, yield a time-series data segment, as follows:

$$x_j = \sum_{n=1}^{N_f} y_n h_{j,n} x_n + w_j. \quad (2)$$

The formula consists of the primary signal sequence and noise w_j , where x_n represents the received modulated signal on the n -th frequency band, and the coefficient $h_{j,n}$ represents the channel gain between the PU and SU- j on the n -th frequency band.

To better extract the spectral features of the observations in the frequency domain with high spectral resolution, the above-mentioned PSD observations need to be collected at a fine granularity, with the resolution of each frequency band being N_w ($N_w \geq 1$). The broadband power spectral density of SU- j is:

$$s_j = \text{FT}(\text{Corr}(x_j)) \in \mathbb{R}^{N_w N_f}, \quad (3)$$

where $\text{FT}(\cdot)$ represents the Fourier transform, and $\text{Corr}(\cdot)$ represents the autocorrelation function. Since this study divides the spectrum pool evenly into a frequency bands, each frequency band has the same bandwidth, and the broadband PSD observation s_j can be expressed as $s_j = [s_{j,1}, \dots, s_{j,n}, \dots, s_{j,N_f}]$, where $s_{j,n} \in \mathbb{R}^{N_w}$, $n = 1, 2, \dots, N_f$.

3.2 Deep Learning-Based Wideband Spectrum Sensing

First, the mathematical representation of the spectrum sensing problem needs to be abstracted. The spectrum occupancy detection of a single secondary user on a single frequency band is a binary hypothesis testing problem, where the frequency band has only two possible states: occupied (1) or idle (0). For the PSD observation $s_{j,n} \in \mathbb{R}^{N_w}$ of secondary user SU- j on the n -th frequency band, the spectrum sensing problem can be expressed as a function:

$$f_n(s_{j,n}, W_j) \in [0, 1], \quad (4)$$

where $W = [W_1, \dots, W_{N_w}]$ represents the parameters of the output layer neurons, which need to be optimized through training to find the optimal value W^* .

In the context of broadband spectrum sensing, it can be represented as a function $\tilde{y} = f(s; W)$, where the input data $s = [s_1, \dots, s_{N_w}]$ yields a decision statistic vector $\tilde{y} \in \mathbb{R}^{N_f}$ for N_f frequency bands. Given an appropriately chosen loss function $L(\tilde{y}, y)$,

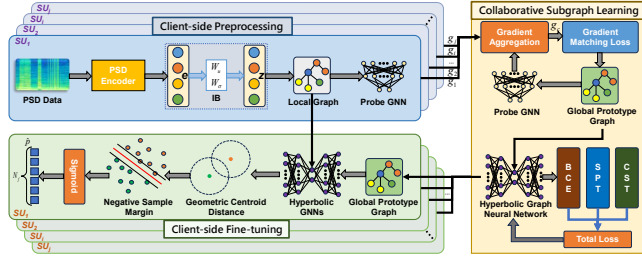


Figure 2: Framework of the proposed method

the parameters W of the deep model can be trained on a dataset D with labeled outputs y , as follows:

$$\arg \min_W \sum_{\{s,y\} \in D} L(f(s; W), y), \quad (5)$$

where $y \in \{0, 1\}^{N_f}$. In this scenario, traditional deep learning-based spectrum sensing models perform pattern matching among 2^{N_f} spectrum occupancy patterns, leading to an exponential increase in the search space. To maintain model performance under these conditions, higher model complexity is required, and the demand for training data also significantly increases. This paper employs graph neural networks for training, which avoids the 2^{N_f} -classification problem with exponential complexity, thereby enhancing training efficiency and scalability.

4 System Model

As shown in Figure 2, this paper aims to develop a novel spectrum sensing framework based on collaborative subgraph learning. It models the spectrum sensing data of distributed secondary users as narrowband spectrum subgraphs and conducts collaborative subgraph training to obtain the full wideband spectrum graph for wideband spectrum occupancy detection. In this section, the method of modeling heterogeneous spectrum graphs using spectrum sensing data is first introduced, which is the basis of the method proposed in this paper. Then, this section will provide a detailed introduction to the spectrum sensing framework based on collaborative subgraph learning.

4.1 Local Processing and Graph Construction at the Client

In the framework proposed in this study, each secondary user first needs to convert its locally observed power spectral density (PSD) data into a graph structure for subsequent graph neural network modeling and collaborative learning. The main objectives of this step are: on the one hand, to extract low-dimensional features that can characterize the spectrum occupancy pattern from the original PSD sequence; on the other hand, to construct a local graph by utilizing the structural relationships within the spectrum, so that local observation data can be input into the subsequent collaborative learning stage in a unified graph representation form, thereby providing a basis for the learning of the wideband spectrum graph. The entire process can be divided into three key links: node construction and adjacency relationship definition, node feature encoding, and information bottleneck compression.

Modeling spectrum data into a graph is the basis for using graph neural networks for spectrum occupancy detection. Previously, the spectrum graph modeling methods used by researchers were mostly relatively rough, which calculated the correlation between PSD observations of different frequency bands, such as using the Pearson correlation coefficient:

$$\rho(s_{j,n}, s_{j,m}) = \frac{\text{cov}(s_{j,n}, s_{j,m})}{\sigma(s_{j,n}) \sigma(s_{j,m})}, n, m \in \{0, 1, \dots, N_f\}, \quad (6)$$

where $s_{j,n}$ and $s_{j,m}$ represent the PSD measurement values of SU- j on the n -th and m -th frequency bands respectively, ρ represents the Pearson correlation coefficient, $\text{cov}(\cdot, \cdot)$ represents the covariance function, and σ represents the standard deviation. When the correlation coefficient is higher than the set correlation threshold, the correlation is set to 1; otherwise, it is set to 0. However, in modern wireless communication systems, factors such as power leakage, discontinuous channel allocation, and carrier aggregation lead to significant heterogeneity in the inter-band correlation of spectrum occupancy patterns. For instance, power leakage and discontinuous channel allocation result in a positive correlation between the PSD measurement variations of different frequency bands.

To better capture the inter-band correlations caused by different reasons, this paper first performs feature extraction on the PSD vector $s_{j,n}$ of each frequency band. Specifically, a one-dimensional convolutional neural network is first used to capture the local patterns of the original PSD signal. Then global average pooling is used to compress time-domain information, and finally, a multi-layer perceptron is connected to obtain an intermediate feature representation with a fixed dimension:

$$e_{j,n} = \text{Enc}(s_{j,n}) \in \mathbb{R}^{d_e}. \quad (7)$$

After concatenating the features of all frequency bands, an intermediate feature matrix $E_j \in \mathbb{R}^{N_f \times d_e}$ can be obtained. This process ensures that the representation extracted from the original spectrum has good discriminability while compressing the dimension of the original signal. However, directly using intermediate features may introduce redundant information, affecting the robustness of the model in low SNR ratio scenarios. For this reason, a variational information bottleneck module is introduced to further compress the features following feature extraction. Specifically, the intermediate feature $e_{j,n}$ output by the encoder is mapped to the mean and variance of the latent distribution:

$$\begin{aligned} \mu_{j,n} &= W_\mu \cdot e_{j,n} + b_\mu, \\ \log \sigma_{j,n}^2 &= W_\sigma \cdot e_{j,n} + b_\sigma. \end{aligned} \quad (8)$$

During the training process, the reparameterization trick is employed for sampling the latent representation from the Gaussian distribution:

$$z_{j,n} = \mu_{j,n} + \sigma_{j,n} \odot \varepsilon, \varepsilon \sim \mathcal{N}(0, I), \quad (9)$$

thus obtaining more compact node features with information constraints. It is worth noting that in the inference stage, to ensure the stability of the output, the mean $\mu_{j,n}$ is directly used as the deterministic representation to ensure output stability. Finally, the latent representations of all frequency bands form a matrix:

$$Z_j = [z_{j,1}, z_{j,2}, \dots, z_{j,N_f}]^T \in \mathbb{R}^{N_f \times d_z}. \quad (10)$$

In the node construction stage, each frequency band observable to the SU is regarded as a graph node. Assuming that the j -th SU can observe N_f frequency bands at the current moment, the node set can be expressed as $\{1, 2, \dots, N_f\}$. To capture the local continuity of the spectrum in the frequency dimension, an inherent correlation is assumed to exist between adjacent frequency bands. Therefore, in the adjacency matrix A_j of the graph, when two frequency bands are adjacent in frequency, the corresponding element is set to 1; otherwise, it is set to 0. At the same time, to ensure the numerical stability of graph convolution, self-loops are added to the adjacency matrix, and the symmetric normalization method is used to obtain the final propagation matrix:

$$\tilde{A}_j = D_j^{-\frac{1}{2}}(A_j + I)D_j^{-\frac{1}{2}}, \quad (11)$$

where D_j is the degree matrix of $A_j + I$. This adjacency matrix not only reflects the structural relationship between frequency bands but also provides normalized weights for subsequent graph convolution propagation.

In summary, each client finally outputs a local graph, which is formally expressed as:

$$G_j = (\tilde{A}_j, Z_j, Y_j, M_j), \quad (12)$$

where $\tilde{A}_j \in \mathbb{R}^{N_f \times N_f}$ is the symmetric normalized adjacency, $Z_j \in \mathbb{R}^{N_f \times d_z}$ is the node feature obtained through the encoder and the information bottleneck, $Y_j \in \{0, 1\}^{N_f}$ is the node label (0 for idle, 1 for occupied), and $M_j \in \{0, 1\}^{N_f}$ is the observable mask. The construction of the local spectrum subgraph provides basic input for subsequent collaborative training and the generation of the full wideband spectrum graph.

4.2 Collaborative Subgraph Learning

After the local processing in the previous stage, the model obtains the local spectrum subgraph converted from the power spectral density snapshot of each secondary user. The goal of the model in this stage is to construct a compact global spectrum graph $\mathcal{G}'$ through collaborative subgraph learning without centralizing the original data.

$$\mathcal{G}' = (A', X', Y'), \quad X' \in \mathbb{R}^{N' \times d_z}, A' \in \mathbb{R}^{N' \times N'}, \quad (13)$$

where $N' \ll N_f$. Each node of this global graph can be regarded as a prototype band, and its features and edge structure are learned through optimization, so that the parameter gradient generated by training with $\mathcal{G}'$ on the isomorphic graph network is as close as possible to the aggregated gradient during the joint training of each client's local graph.

To capture the intrinsic topological features of the distributed subgraphs without overfitting to specific model parameters, A random probe ensemble strategy was introduced. Instead of relying on a single fixed network, we utilize a set of randomized probe networks f_θ . The probe networks serve as random projection functions that map the graph structure into the gradient space. According to the Johnson-Lindenstrauss lemma, matching the gradients across multiple random projections ensures that the reconstructed global graph $\mathcal{G}'$ preserves the geometric properties of the underlying data distribution with high probability, even without transmitting the raw data.

For the convenience of subsequent derivation, first define the available sample set and class-conditional subsets on client j :

$$\begin{aligned} \mathcal{V}_j &= \{n \mid m_{j,n} = 1\}, \\ \mathcal{V}_{j,c} &= \{n \in \mathcal{V}_j \mid y_{j,n} = c\}, \quad c \in \{0, 1\}, \end{aligned} \quad (14)$$

where n is the index of the observable frequency band of the client, $m_{j,n} \in \{0, 1\}$ is an observable mask ($m_{j,n} = 1$ means that the frequency band is visible to client j in the current frame, $m_{j,n} = 0$ means it is invisible or missing and does not participate in any local risk and gradient calculation), $y_{j,n} \in \{0, 1\}$ is the node label (0 for idle, 1 for occupied). The set $\mathcal{V}_j$ only includes visible frequency bands, and $\mathcal{V}_{j,c}$ further stratifies by category c within the visible frequency bands. Obviously, $\mathcal{V}_j = \mathcal{V}_{j,0} \cup \mathcal{V}_{j,1}$ and $\mathcal{V}_{j,0} \cap \mathcal{V}_{j,1} = \emptyset$.

In terms of statistics, let the local class count and total count be

$$n_j(c) = |\mathcal{V}_{j,c}|, \quad n_j = |\mathcal{V}_j|, \quad (15)$$

and define the local class prior:

$$\pi_j(c) = \frac{n_j(c)}{\max(1, n_j)}, \quad c \in \{0, 1\}, \quad (16)$$

where $\max(1, n_j)$ is used to avoid the degenerate case of $n_j = 0$. To characterize the observation quality, the model introduces a weight $w_{j,n} \in (0, 1]$ for each node according to the local signal-to-noise ratio and energy stability, and the corresponding weighted class effective sample size is:

$$\tilde{n}_j(c) = \sum_{n \in \mathcal{V}_j} \mathbf{1}[y_{j,n} = c] w_{j,n}, \quad (17)$$

where $\mathbf{1}[\cdot]$ is an indicator function. The server will use $\tilde{n}_j(c)$ as the weight during subsequent aggregation.

The input of client j is the local graph $(\tilde{A}_j, Z_j, Y_j, M_j)$ from the previous stage. Among them, $\tilde{A}_j \in \mathbb{R}^{N_f \times N_f}$ is a symmetric normalized adjacency (obtained by connecting adjacent frequency bands and normalizing with self-loops), and $Z_j \in \mathbb{R}^{N_f \times d_z}$ is the node feature obtained through the encoder and the information bottleneck; Y_j and M_j are labels and masks respectively. Let the probe graph network isomorphic to the subsequent stage be f_θ , which takes $(\tilde{A}_j, Z_j)$ as input and outputs the prediction probability vector of each node belonging to the occupied class:

$$\begin{aligned} \hat{\mathbf{p}}_j &= f_\theta(\tilde{A}_j, Z_j) \in (0, 1)^{N_f}, \\ \hat{\mathbf{p}}_{j,n} &= [\hat{\mathbf{p}}_j]_n, \end{aligned} \quad (18)$$

where θ is the parameter tensor set of the probe network, including all linear layer weights, biases, etc. The probe network is only used to generate comparable gradient signals in this stage, and the hyperbolic GNN will be used for real task learning in subsequent stages.

In terms of constructing the local empirical risk, this paper adopts a binary cross-entropy decomposed by category, and accumulates only on the corresponding visible class subsets; if there are node weights, weighted summation and weight normalization are performed:

$$\begin{aligned} \mathcal{L}_{j,c} &= \frac{1}{\max(\epsilon, \sum_{n \in \mathcal{V}_{j,c}} w_{j,n})} \sum_{n \in \mathcal{V}_{j,c}} w_{j,n} \text{BCE}(\hat{\mathbf{p}}_{j,n}, y_{j,n}), \\ \text{BCE}(\hat{p}, y) &= -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})], \end{aligned} \quad (19)$$

where $\epsilon > 0$ is a numerical stability constant, $c \in \{0, 1\}$. This definition ensures that invisible frequency bands do not enter any risk terms, and the risks of different categories are comparable across different clients. From this, the class-conditional gradient of the probe parameters is obtained:

$$g_{j,c} = \nabla_{\theta} \mathcal{L}_{j,c} \in \mathbb{R}^{\dim(\theta)}. \quad (20)$$

The content uploaded by client j to the server is $\{(g_{j,c}, \omega_{j,c})\}_{c \in \{0,1\}}$, where $\omega_{j,c}$ takes $n_j(c)$ or $\tilde{n}_j(c)$. The upload does not contain any original PSD, feature matrix or adjacency matrix, which meets the requirements of privacy and controllable communication overhead.

The server robustly aggregates the received gradients in the category dimension to obtain the global learning direction for category c :

$$g_c = \text{Agg}(\{(g_{j,c}, \omega_{j,c})\}_j), \quad (21)$$

where aggregator $\text{Agg}(\cdot)$ can adopt robust strategies such as coordinate median, truncated mean or weighted mean according to weights; $\omega_{j,c}$ is used as a weight to characterize the effective sample size and quality of the client in category c . The obtained g_c approximates the parameter update direction when training the probe network f_{θ} on the cross-client real distribution.

Next, the server side uses a small-scale global spectrum graph $\mathcal{G}' = (A', X', Y')$ as a learnable "prototyped" graph representation. Here, $X' \in \mathbb{R}^{N' \times d_z}$ is the prototype node feature matrix and a learnable parameter; $A' \in [0, 1]^{N' \times N'}$ is a continuous edge-weighted adjacency matrix, which is generated by an edge generator based on the Pearson correlation coefficient; $Y' \in \{0, 1\}^{N'}$ is the prototype label. The node features are initialized as a small-scale Gaussian distribution $x'_i \sim \mathcal{N}(0, \sigma_x^2 I)$ (where σ_x is usually set to 0.02), and the label quota N'_c is allocated according to the relative frequency of $\sum_j \omega_{j,c}$. For edge generation, a symmetric MLP scorer is used: for any i, j , a concatenated input is constructed as follows:

$$u_{ij} = [x'_i, x'_j, \|x'_i - x'_j\|_2^2] \in \mathbb{R}^{2d_z+1}. \quad (22)$$

Then, the score is calculated through a multi-layer perceptron $\Phi_{\psi}(\cdot)$, and the undirected weighted adjacency is obtained via the Sigmoid function:

$$\begin{aligned} S_{ij} &= \Phi_{\psi}(u_{ij}), \\ A'_{ij} &= \frac{1}{2}(\sigma(S_{ij}) + \sigma(S_{ji})), \\ \sigma(t) &= \frac{1}{1 + e^{-t}}. \end{aligned} \quad (23)$$

After each inner-layer update, symmetric normalization is applied to obtain the propagation matrix:

$$\begin{aligned} \tilde{A}' &= D'^{-\frac{1}{2}}(A' + I)D'^{-\frac{1}{2}}, \\ D'_{ii} &= \sum_k (A'_{ik} + \delta_{ik}), \end{aligned} \quad (24)$$

which is used for the forward propagation of the probe network on the global graph.

To make the learning signal of the global graph on the probe network reproduce the real cross-client learning signal, the server constructs a class-conditional risk on $\mathcal{G}'$ that is consistent with that on the clients, and computes the gradient with respect to the probe

Table 1: THE MAIN PARAMETERS OF DATASET AND EXPERIMENT

Parameters	Values
The dimensionality of samples	64*20
Original learning rate	0.05
Learning rate decay	0.2
Batch size	50
Maximum epoches	3000
Dataset split ratio	8:2

parameters to obtain the "synthetic gradient":

$$\begin{aligned} \mathcal{L}'_c &= \frac{1}{|\mathcal{V}'_c|} \sum_{i \in \mathcal{V}'_c} \text{BCE}(\hat{p}'_i, y'_i), \\ \hat{p}'_i &= f_{\theta}(\tilde{A}', X')_i, \\ g'_c &= \nabla_{\theta} \mathcal{L}'_c, \end{aligned} \quad (25)$$

where $\mathcal{V}'_c = \{i \mid y'_i = c\}$ is the class subset of the global prototype graph. Here, θ is treated as a constant and only used for direction comparison; the parameters that are actually optimized are X' and ψ .

To align the learning directions of the two classes, a unit-norm normalization operator $\mathcal{N}(G) = G/\|G\|_2$ is used to compare the direction consistency, resulting in the gradient matching loss:

$$\mathcal{L}_{\text{GM}} = \sum_{c \in \{0,1\}} \|\mathcal{N}(g'_c) - \mathcal{N}(g_c)\|_2^2. \quad (26)$$

To avoid obtaining an excessively dense or abnormally structured graph, sparsity and degree distribution regularization are introduced:

$$\begin{aligned} \mathcal{L}_{\text{sp}} &= \|A'\|_1, \\ \mathcal{L}_{\text{deg}} &= \text{Var}(\text{deg}(A')), \\ \text{deg}_i(A') &= \sum_j A'_{ij}. \end{aligned} \quad (27)$$

The overall objective function of this stage is:

$$\min_{X', \psi} \mathcal{L}_{\text{CSL}} = \mathcal{L}_{\text{GM}} + \lambda_{\text{sp}} \mathcal{L}_{\text{sp}} + \lambda_{\text{deg}} \mathcal{L}_{\text{deg}}, \quad (28)$$

where $\lambda_{\text{sp}}, \lambda_{\text{deg}} > 0$ are regularization weights. This objective function expresses the following requirement: under the isomorphic probe network f_{θ} , the update direction obtained by training with the global prototype graph $\mathcal{G}'$ should be as close as possible to the update direction of the real cross-client distribution. Details of subsequent pre-training and local fine-tuning are provided in Appendices A and B.

5 Experiments and Discussion

This section introduces the real dataset and simulated dataset used in the experiment, as well as the experimental settings and results. It also conducts performance comparison with multiple benchmark methods, along with ablation experiments and result analysis. Details of the experimental settings and the datasets used are provided in Appendices C and D.

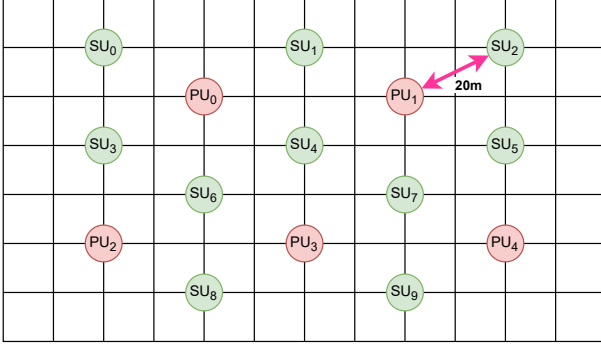


Figure 3: Locations of PUs and SUs in the CR system

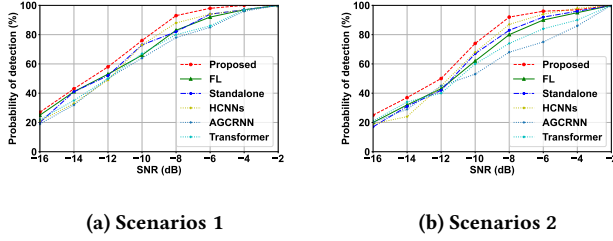


Figure 4: Figure 4 shows the PD of different SNR Conditions for Scenarios 1-2

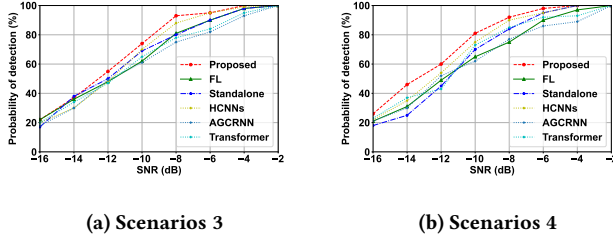


Figure 5: Figure 5 shows the PD of different SNR Conditions for Scenarios 3-4

Table 2: Band Utilization of PUs

PU index	Assigned bands	Modulation scheme
0	1,9,12	MSK
1	2,8,11	4FSK
2	0,3,5,10	BPSK
3	6,13,17	MSK
4	4,7,14	16QAM

5.1 Performance Evaluation

This section compares the performance of the proposed method with baseline methods across the four previously defined experimental scenarios. First, the clustering of frequency band correlations is visualized. Then, the training convergence under different

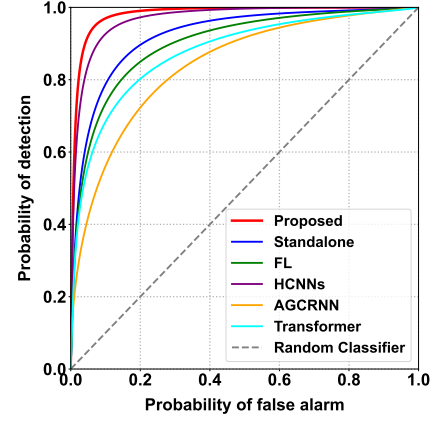


Figure 6: ROC of different models in real world dataset

SNR environments is evaluated, and the ROC curves and detection probability (PD) curves for different methods are plotted.

To ensure the reliability of the data and the rigor of the conclusions, all experimental indicators are obtained by taking the average value and calculating the standard deviation after multiple independent repeated experiments, each group of experiments is repeated 10 times.

As shown in Table 3, the experimental results show that the proposed collaborative spectrum sensing model based on hyperbolic GNN and collaborative subgraph learning significantly outperforms existing benchmark models in terms of low model complexity and computational cost. Although HCNns, standalone models, and Federated Learning frameworks perform well in spectrum sensing, they have higher training time and memory consumption. After multiple experiments and averaging, their training duration is 40% - 80% longer than that of the proposed method, and their memory usage is 10% - 35% higher. In contrast, the proposed method significantly reduces computational and memory costs by introducing hyperbolic geometry and graph representation compression. Verified by multiple experiments, the average computational cost is reduced by approximately 40%, the memory overhead is reduced by about 25%, the training time and inference time are shortened by about 40% and 42% respectively, and the standard deviation of the experimental results is controlled within 5%, with good stability. This is crucial for real-time spectrum sensing tasks.

Scenario 1: As shown in Table 4 and Figure 4, the proposed method achieves the highest detection accuracy across all three SNR conditions in Scenario 1, reaching 97.49% at -5dB, 84.73% at -10dB, and 62.46% at -15dB. It outperforms all baseline methods, with particularly significant advantages over AGCRNN and Transformer models. The performance gap is most pronounced at lower SNR values, demonstrating the method's robustness in challenging signal conditions.

Scenario 2: As shown in Table 4 and Figure 4, under reduced training data conditions, the proposed method maintains its performance advantage with accuracies of 96.13% (-5dB), 83.57% (-10dB), and 62.18% (-15dB). While all methods experience performance degradation compared to Scenario 1, the standalone approach shows the

Table 3: Comparison of Computational Complexity and Efficiency

Model	Params (M)	Training Time (h)	Inference Time (ms)	Memory Usage (GB)
HCNNs	0.42	4.52	1.23	4.31
Standalone	0.57	5.20	1.41	5.13
FL	0.68	5.95	1.73	5.35
AGCRNN	0.54	5.85	2.45	5.27
Transformer	1.27	6.82	3.18	13.58
Proposed Model	0.35	3.24	0.98	3.95

Table 4: Detection Accuracy (%) Under Different SNR Conditions for Scenarios 1-2

Method	Scenario 1			Scenario 2		
	-5dB	-10dB	-15dB	-5dB	-10dB	-15dB
FL	95.27	78.54	52.31	88.39	70.42	54.32
Standalone	94.63	76.21	54.84	94.39	80.32	54.76
HCNNs	96.35	82.39	60.29	95.37	82.52	60.29
Agcrnn	92.45	78.93	58.57	91.87	77.36	57.28
Transformer	93.21	79.31	59.86	93.56	78.39	58.83
Proposed	97.49	84.73	62.46	96.13	83.57	62.18

Table 5: Detection Accuracy (%) Under Different SNR Conditions for Scenarios 3-4

Method	Scenario 3			Scenario 4		
	-5dB	-10dB	-15dB	-5dB	-10dB	-15dB
FL	92.13	74.21	48.42	84.57	68.53	51.02
Standalone	91.39	73.04	50.44	90.63	77.25	50.52
HCNNs	93.24	78.82	56.21	90.57	79.46	55.92
Agcrnn	88.73	75.41	54.72	87.39	73.79	53.13
Transformer	90.16	76.24	55.62	88.47	75.62	55.49
Proposed	95.82	81.35	60.77	94.96	79.42	59.83

smallest drop in the -5dB condition but suffers significant performance loss at lower SNRs. This highlights the particular challenge of data scarcity in low SNR environments.

Scenario 3: As shown in Table 5 and Figure 5, in the channel parameter variation scenario, the proposed method continues to deliver the best performance with 95.82% (-5dB), 81.35% (-10dB), and 60.77% (-15dB) accuracy. The performance ranking among methods remains largely consistent with previous scenarios, with HCNNs maintaining the second-best position. This demonstrates the method's adaptability to changing channel conditions.

Scenario 4: As shown in Table 5 and Figure 5, facing unknown modulation types, the proposed method achieves 94.96% (-5dB), 79.42% (-10dB), and 59.83% (-15dB) accuracy, maintaining its leading position. The performance gaps between methods are generally consistent with those observed in Scenario 3, suggesting that modulation uncertainty affects all methods to a similar degree, without altering their relative performance rankings.

Real world dataset: The ROC curves in Figure 6 demonstrate the proposed method's significant advantage in real-world spectrum sensing scenarios. The proposed approach achieves the highest true positive rates across all false positive rate levels, with its curve positioned consistently above all baseline models. Among the compared methods, HCNNs deliver the second-best performance, showing it's effectiveness in handling real-world spectrum data. The AGCRNN and Transformer models demonstrate competitive but somewhat inferior performance compared to HCNNs. Federated learning exhibits the steepest initial curve, suggesting reasonable detection capability but limited overall performance. The standalone method demonstrates the weakest performance, highlighting the limitations of isolated learning without collaborative information sharing. These results validate the proposed method's practical effectiveness in handling complex real-world signal environments.

6 Conclusion

This paper proposes a collaborative subgraph learning framework for wideband spectrum occupancy detection, designed to address partial observation issues in practical scenarios. The framework employs distributed graph neural networks to extract frequency band features and inter-band correlations, models the observed narrow-band segments as subgraphs, and through collaborative training and aggregation optimization, constructs the entire wideband pool as a complete graph, thereby significantly enhancing overall spectrum sensing performance. Experimental results demonstrate that the framework outperforms existing baseline methods in small-data environments and effectively addresses the dual challenges of partial observation and data scarcity, making it an ideal solution for resource-constrained cognitive radio applications.

Acknowledgments

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A Pre-training of Hyperbolic GNN

In this stage, the server trains a hyperbolic graph neural network on the fixed $\mathcal{G}'$ to extract the structured prior shared across clients; after training, the network parameters and geometric class centers are output and distributed to each client for local fine-tuning and inference.

For unified expression, the curvature constant $c > 0$ is fixed (for example, $c = 0.5$), and the Poincaré ball model $\mathbb{D}_c^d = \{x \in \mathbb{R}^d : \|x\| < 1/\sqrt{c}\}$ is adopted. In this model, the exponential/logarithmic mapping of the origin is:

$$\begin{aligned} \exp_0^c(v) &= \begin{cases} \tanh(\sqrt{c} \|v\|) \frac{v}{\sqrt{c} \|v\|}, & \|v\| > 0 \\ 0, & \|v\| = 0 \end{cases}, \\ \log_0^c(x) &= \begin{cases} \frac{1}{\sqrt{c}} \operatorname{arctanh}(\sqrt{c} \|x\|) \frac{x}{\|x\|}, & \|x\| > 0 \\ 0, & \|x\| = 0 \end{cases}. \end{aligned} \quad (29)$$

The geodesic distance between two points $x, y \in \mathbb{D}_c^d$ is:

$$d_{\mathbb{H}}(x, y) = \operatorname{arcosh} \left(1 + \frac{2\|x - y\|^2}{(1 - \|x\|^2)(1 - \|y\|^2)} \right). \quad (30)$$

To stabilize the message passing on the graph, the symmetric normalized propagation matrix is fixed throughout the training process:

$$\begin{aligned}\tilde{A}' &= D'^{-\frac{1}{2}}(A'_{\text{sp}} + I)D'^{-\frac{1}{2}}, \\ D'_{ii} &= \sum_k (A'_{\text{sp}}[i, k] + \delta_{ik}).\end{aligned}\quad (31)$$

In the above formula, I is the identity matrix, δ_{ik} is the Kronecker symbol; the role of $\tilde{A}'$ is to provide normalized weights for each layer of propagation.

The model takes X' as input and first embeds the Euclidean features into the hyperbolic space through $\exp_0^c$. The initial representation is denoted as:

$$H^{(0)} = \exp_0^c(X') \in \mathbb{D}_c^{N' \times d_z}. \quad (32)$$

Subsequently, two-layer message passing of "tangent space linearity-hyperbolic space writing back" is performed. Let the hidden dimension and output dimension be d_h and d_o respectively, and the weight matrices be $W^{(0)} \in \mathbb{R}^{d_z \times d_h}$ and $W^{(1)} \in \mathbb{R}^{d_h \times d_o}$. Then the update of the l -th layer ($l = 0, 1$) is written as:

$$H^{(l+1)} = \exp_0^c\left(\phi(\tilde{A}' \log_0^c(H^{(l)}) W^{(l)})\right). \quad (33)$$

Specifically, $\log_0^c(H^{(l)})$ maps the hyperbolic representation of the previous layer back to the tangent space at the origin; $\tilde{A}' \cdot W^{(l)}$ completes graph propagation and linear transformation in the tangent space; $\phi(\cdot)$ introduces nonlinearity; $\exp_0^c(\cdot)$ writes the result back to the hyperbolic space. After two layers of propagation, the node representation $H^{(2)} = \{h_i\}_{i=1}^{N'} \subset \mathbb{D}_c^{d_o}$ is obtained. To avoid numerical instability, after each new hyperbolic vector is obtained, if $\|h_i'\| \geq 1/\sqrt{c} - \varepsilon$ ($\varepsilon \approx 10^{-5}$), it is scaled radially to within the radius $1/\sqrt{c} - \varepsilon$; the same processing is also performed on the class centers involved later.

Based on the obtained $H^{(2)}$ and the label Y' , the Fréchet means (geometric centers) of the two classes are estimated in the hyperbolic space, respectively:

$$c'_y = \arg \min_{p \in \mathbb{D}_c^{d_o}} \sum_{i: y_i=y} d_{\mathbb{H}}^2(h'_i, p), \quad y \in \{0, 1\}. \quad (34)$$

In actual calculation, the iterative algorithm of Karcher mean can be used: the weighted average of $\{\log_p(h'_i)\}$ is performed in the tangent space of the current center p , and then returned to the hyperbolic space with $\exp_p(\cdot)$; for simplicity of implementation, this paper defaults to approximation at the origin, and the above projection constraint is applied to c'_y after each outer loop. A classification score is constructed based on the distance difference to the two class centers, and the 'occupation' probability is obtained using Sigmoid:

$$\begin{aligned}\text{logit}_i &= \tau[d_{\mathbb{H}}(h'_i, c'_0) - d_{\mathbb{H}}(h'_i, c'_1)], \\ \hat{p}'_i &= \sigma(\text{logit}_i),\end{aligned}\quad (35)$$

where $\tau > 0$ is the temperature coefficient, and $\sigma(\cdot)$ is Sigmoid. The geometric meaning of logit is "which center it is close to", and τ controls its steepness.

The training objective takes into account discriminability, structural fidelity and inter-class geometric separation. First, the node-level supervision uses binary cross-entropy:

$$\mathcal{L}_{\text{cls}} = \frac{1}{N'} \sum_{i=1}^{N'} \left[-y'_i \log \hat{p}'_i - (1 - y'_i) \log(1 - \hat{p}'_i) \right], \quad (36)$$

which directly improves the separability of prototype nodes. Secondly, to retain the previously learned connection geometry and prevent the representation from collapsing into a fully connected structure, a structure-preserving term of "adjacency smoothing + negative sample margin" is introduced. Let the edge set $E' = \{(i, j) \mid A'_{\text{sp}}[i, j] > 0\}$, and sample the negative sample pair set $\mathcal{N}^-$ from non-adjacent or heterogeneous nodes, then:

$$\mathcal{L}_{\text{str}} = \frac{1}{|E'|} \sum_{(i,j) \in E'} d_{\mathbb{H}}^2(h'_i, h'_j) + \frac{1}{|\mathcal{N}^-|} \sum_{(i,k) \in \mathcal{N}^-} [m_s - d_{\mathbb{H}}(h'_i, h'_k)]_+, \quad (37)$$

where $[t]_+ = \max(0, t)$, $m_s > 0$ is the margin hyperparameter; the first term encourages the connected prototypes to be close in the geometric space, and the second term requires the distance between non-adjacent or heterogeneous pairs to be at least greater than m_s . In addition, to pull apart the global boundary between the two classes geometrically, a class center separation term is introduced:

$$\mathcal{L}_{\text{cen}} = -d_{\mathbb{H}}(c'_0, c'_1), \quad (38)$$

that is, to maximize the geodesic distance between the two class centers. Combining them together, the total loss is obtained:

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{cls}} + \alpha \mathcal{L}_{\text{str}} + \beta \mathcal{L}_{\text{cen}}, \quad (39)$$

where $\alpha, \beta > 0$ are weight coefficients. The roles of the three terms correspond to "point separability", "edge fidelity and negative pair margin", and "global geometric boundary" respectively, which together ensure that the model not only follows the structural prior but also has a good discriminative margin.

The optimization is only performed on the parameters and temperature of the hyperbolic GNN, denoted as:

$$\begin{aligned}\theta^{\text{hyp}} &= \{W^{(0)}, W^{(1)}, \tau\}, \\ \min_{\theta^{\text{hyp}}} \mathcal{L}_{\text{pre}}(\theta^{\text{hyp}} \mid A'_{\text{sp}}, X', Y', c).\end{aligned}\quad (40)$$

The global graph (A'_{sp}, X', Y') and the curvature c are fixed in this stage. In implementation, the whole graph is trained (the whole $\mathcal{G}'$ is a batch), and the optimizer uses Adam (learning rate 10^{-3} , weight decay 10^{-4}). After each round of training, all h'_i and c'_y are projected back into the ball with radius $1/\sqrt{c} - \varepsilon$, and τ is clipped to a reasonable interval (such as $[0.5, 5]$) to avoid numerical instability. It is recommended to sample 5–10 non-adjacent or heterogeneous pairs at each node for negative samples, so that $|\mathcal{N}^-|$ and $|E'|$ are of the same order of magnitude, thereby stabilizing the gradient of the structure term.

After the training converges, the server produces two types of results: one is the network parameters and temperature:

$$\theta^{\text{hyp}} = \{W^{(0)}, W^{(1)}, \tau\}, \quad (41)$$

the other is the category geometric centers (c'_0, c'_1) in the hyperbolic space. Together with the curvature constant c , they form a parameter package, which is distributed to each client and used as a unified initialization and geometric anchor point for local fine-tuning. On

the client side, the node features will be embedded into the hyperbolic space with the same curvature, and the occupation probability of each local frequency band is obtained through the geodesic distance difference from (c'_0, c'_1) , and then light-weight fine-tuning is performed in combination with local supervision and regularization. Since $N' \ll N_f$, the calculation and storage overhead of this pre-training stage is small, and the structured knowledge across clients is transmitted in a compact manner, providing a stable starting point for subsequent personalized adaptation.

B Personalized Fine-tuning and Inference

The goal of this stage is to use the local graph to perform light-weight fine-tuning on the above pre-trained model without sharing the original data, so as to obtain a spectrum detector that is more suitable for the local noise environment and observation coverage. After completing the fine-tuning, the client will use the same network structure for online inference.

For the sake of unified description, the single-frame local graph of client j is still denoted as $\mathcal{G}_j = (\tilde{A}_j, Z_j, Y_j, M_j)$. Among them, $\tilde{A}_j = D_j^{-\frac{1}{2}}(A_j + I)D_j^{-\frac{1}{2}}$ is the symmetric normalized propagation matrix of the adjacent frequency band graph; $Z_j \in \mathbb{R}^{N_f \times d_z}$ is the output of VIB; $Y_j \in \{0, 1\}^{N_f}$ and $M_j \in \{0, 1\}^{N_f}$ are the node labels and observable masks respectively. During training, several local snapshots are processed in batches, and each snapshot can form a $\mathcal{G}_j$.

Fine-tuning takes the pre-trained weights as initialization and the distributed geometric centers as anchors. Specifically, first, the Euclidean features of each node are embedded into the Poincaré ball with curvature c , and the following result is obtained:

$$H^{(0)} = \exp_0^c(Z_j) \in \mathbb{D}_c^{N_f \times d_z}, \quad (42)$$

where $\exp_0^c(\cdot)$ is the exponential mapping of the origin; if $\|H_n^{(0)}\| \geq 1/\sqrt{c} - \varepsilon$ ($\varepsilon \approx 10^{-5}$), it is scaled back radially to within the radius $1/\sqrt{c} - \varepsilon$ to ensure numerical stability. Subsequently, the two-layer message passing involves tangent space linearity and hyperbolic space writing back, which is isomorphic to the pre-training stage, is adopted:

$$H^{(l+1)} = \exp_0^c\left(\phi(\tilde{A}_j \log_0^c(H^{(l)}) W^{(l)})\right), \quad l = 0, 1, \quad (43)$$

where $\log_0^c(\cdot)$ is the logarithmic mapping of the origin, $W^{(0)}, W^{(1)}$ are initialized from θ^{hyp} and updated locally, and ϕ takes ReLU. After two layers of propagation, the hyperbolic representation $H^{(2)} = \{h_{j,n}\}_{n=1}^{N_f} \subset \mathbb{D}_c^{d_o}$ of each frequency band is obtained. The classification score follows the geometric discrimination form of Stage 1. The geodesic distance difference to the two class centers is used as the logit, and the "occupation" probability is obtained through Sigmoid:

$$\begin{aligned} \text{logit}_{j,n} &= \tau[d_{\mathbb{H}}(h_{j,n}, c'_0) - d_{\mathbb{H}}(h_{j,n}, c'_1)], \\ \hat{p}_{j,n} &= \sigma(\text{logit}_{j,n}) \in (0, 1), \end{aligned} \quad (44)$$

where the initial value of τ is taken from θ^{hyp} and can be updated locally; (c'_0, c'_1) is frozen throughout this stage and used as a unified geometric anchor to avoid uncontrollable drift of the discrimination boundary of each client. The above forward mapping produces

$\{\hat{p}_{j,n}\}$ from $(\tilde{A}_j, Z_j)$ through $\{\exp_0^c, \log_0^c\}$, two-layer graph propagation and geometric scoring, and projects all hyperbolic vectors in necessary places.

This term encourages adjacent frequency bands to stay close in hyperbolic space, thereby retaining local structural priors and suppressing isolated deviations caused by noise. Finally, to enforce the information bottleneck constraint and mitigate the risk of overfitting intermediate representations, we incorporate a KL regularization term into the total objective. This term penalizes the deviation of the node-level latent distributions from a standard normal distribution. By combining these components, the overall objective function for the client is formulated as a weighted sum of the classification loss, the prototype margin loss, the structural smoothing term, and the information bottleneck regularization. In our default setting, fixed hyperparameters are assigned to these terms to balance their contributions and ensure consistent training dynamics across all clients.

After completing the fine-tuning, the client can perform a forward pass on the newly arrived PSD snapshot and directly output the occupation probability and binary decision of the entire frequency band. Since the message passing is completely based on the local $\tilde{A}_j$ and the discrimination boundary is anchored by the geometric centers shared across ends, the model can not only absorb the statistical characteristics within the local domain but also maintain consistency with the global structural prior, thereby achieving stable and efficient wide-band spectrum sensing under conditions of partial observation and low signal-to-noise ratio.

C Simulated Dataset

C.1 Data Generation

This section first describes the method for generating a simulated wideband dataset to address the scarcity of training data for wide-band sensing under partial observation conditions in cognitive radio. For multi-band correlations, this paper primarily considers power leakage and multi-band occupancy. Power leakage is an objective physical phenomenon, while multi-band occupancy is one of the most commonly used dynamic channel allocation strategies.

The narrowband transmission signal x_n on the n -th frequency band is generated as a random message sequence based on predefined modulation schemes. Different modulation schemes can produce different PSD waveforms, thereby endowing each frequency band with distinct intra-band characteristics.

To simulate power leakage between adjacent frequency bands, the main lobe and side lobe PSD of the n -th frequency band need to be represented separately, i.e., $s'_n \in \mathbb{R}^{3N_w}$:

$$s'_n = [s'_{n,L}, s'_{n,M}, s'_{n,R}] \quad (45)$$

where, $s'_{n,M} \in \mathbb{R}^{N_w}$ represents the main lobe PSD, $s'_{n,L} \in \mathbb{R}^{N_w}$ denotes the right side lobe leaked to frequency band $n-1$, and $s'_{n,R} \in \mathbb{R}^{N_w}$ represents the left side lobe leaked to frequency band $n+1$. While frequency band n leaks power to adjacent bands, it is also affected by power leakage from neighboring bands, specifically bands $n-1$ and $n+1$. The PSD value s'_n received by the secondary

user SU- j for frequency band n is calculated as follows:

$$s_n^j = y_n \left| h_n^j \right|^2 s_{n,M} + y_{n-1} \left| h_{n-1}^j \right|^2 s_{n-1,R} + y_{n+1} \left| h_{n+1}^j \right|^2 s_{n+1,L} + w_n, \quad (46)$$

where, y_n represents the occupancy state of frequency band n , h denotes the channel gain of frequency band n , and w represents the additive white noise on this frequency band. During the data generation process, power leakage is simulated by repeatedly applying Equation (51).

For frequency band n , when the secondary user SU- j is far from the PU, the channel gain h will decrease:

$$y_n \left| h_n^j \right|^2 s'_{n,M} \ll s_n^j. \quad (47)$$

Therefore, $s_n^j \Big|_{y_n=1} \approx s_n^j \Big|_{y_n=0}$, which means that the occupancy state of frequency band n cannot be learned from s_n^j . This is also one of the reasons for the occurrence of partial observation scenarios.

C.2 Experimental Setup

Simulation Setup: In the simulation data generation, a multi-band cognitive radio environment consists of $N_f = 15$ frequency bands. For spectrum occupancy detection, the PSD observation for each frequency band are sampled at $N_w = 64$ frequency points. The input dimension of the single PSD data for each secondary user is 64×15 . The CR system includes 5 PUs and 15 SUs, with their locations illustrated in Figure 3. Each secondary user is allocated one or more non-contiguously assigned frequency bands. To simulate shadowing effects reflecting the channel shadow fading between PU and SU $_j$ on frequency band n , the path loss exponent is set to $\alpha = 3.5$, the constant related to antenna characteristics and average attenuation is $\beta = 10^{-3.154}$, and the reference distance is $d_0 = 1$ meter. To simulate the partial observation problem, each SU is configured to only reflect the occupancy status of primary user frequency bands within a 20-meter radius. This study evaluates the performance of different frequency band sensing methods within a SNR range of $[-16, -2]$ dB.

Testing and Training: The experimental environment is configured to ensure reproducibility and robustness. For hardware, all experiments are conducted on a server equipped with an NVIDIA RTX 4090 GPU (24GB VRAM) and an AMD Ryzen 9 7950X CPU (16 cores, 5.7 GHz base clock). The system runs on Ubuntu 22.04 LTS as the operating system, with 32GB of DDR5 RAM for memory-intensive operations. Under partial observation conditions, the experiment is set up with 10 secondary users for multi-band spectrum sensing training. The final occupancy state of each specific frequency band is determined by voting among all secondary users and the global model that can observe that band. This study uses independent learning methods and federated learning methods as comparative experiments. Additionally, clustering visualizations of frequency band correlations are designed, along with ablation experiments that mask heterogeneous graph modeling.

This study designs four different experimental scenarios. In Scenario 1, for each occupancy pattern (totaling $2^{N_{PU}} = 32$ patterns), each secondary user has 100 local training samples. The simulation settings for training and testing data are identical. The remaining

four scenarios modify the sample size and data generation methods based on Scenario 1. In Scenario 2, to simulate the issue of insufficient training data, each secondary user has only 50 samples per occupancy pattern. In Scenario 3, the simulation settings change the path loss from 3.71 to 5.0 to test the model's performance under more challenging channel conditions. In Scenario 4, the experiment modifies the modulation types of the test data to evaluate the model's capability in dynamic spectrum access scenarios. Detailed settings are shown in Table 2.

Baseline DNNs: Given limited research on learning-based collaborative spectrum sensing under distributed partial observations, this study uses prior research and simulations as benchmarks: heterogeneous CNN (HCNN)-based collaborative models, standalone models, and federated learning frameworks. HCNN collaborative model: Each SU's multi-task DNN has 4 convolutional layers + 1 fully connected output layer. The first convolutional layer (shared, 40 filters) is followed by 3 decoupled band-specific layers (8 filters each). The output layer is band-decoupled, with neurons receiving signals from their subnetwork's last convolutional layer for sigmoid output.

Independent learning: Each SU trains a heterogeneous dense CNN (matching HCNN layer neuron counts) on local data to detect observable bands. **Federated learning:** Each SU trains a homogeneous dense CNN (20-band version of the independent model, with 160 neurons in layers 2–4). All DNNs use labels y (all band occupancy) for training; testing uses majority voting for each band. **AGCRNN** is designed for multi-band spectrum prediction. It integrates graph convolutional layers to capture spatial dependencies among frequency bands, gated recurrent units (GRUs) to model temporal dynamics, and an attention mechanism to adaptively focus on relevant temporal states. This combination enables effective learning of spatiotemporal spectrum occupancy patterns under partial observability. **Spectrum Transformer** is built on a core multi-head self-attention mechanism. It is specifically designed to handle the high-dimensional data and inter-band correlations inherent in wideband spectrum sensing. The architecture is capable of simultaneously learning fine-grained features within individual frequency bands and capturing occupancy correlations across different bands.

D Real World Dataset

The real world dataset used in the experiment comes from the experimental platform of Nanjing University of Aeronautics and Astronautics. The platform uses devices such as Universal Software Radio Peripheral (USRP), vector signal generators, spectrum analyzers, and spectrum sensors to collect multi-channel time-frequency power data in the 1300-1700 MHz frequency range. The dataset is collected by measuring power values every 0.1 seconds, and the availability of the channels is labeled based on the measured power values. If the power value exceeds -50 dBm, the channel is considered unavailable. The dataset is split using K-fold cross-validation (K=5), with the data divided into training, validation, and test sets in a 5:1:1 ratio.